Ian Vann

Fair Lawn NJ, 07410
(201) 691 - 4929
https://www.linkedin.com/in/ian-vann/ Icv3@cornell.edu

EDUCATION

Cornell University, College of Engineering, Ithaca, NY **Aug 2023 - May 2027**
Bachelor of Science, Chemical and Biomolecular Engineering, GPA: 3.80

EXPERIENCE

Merck, Rahway, NJ, *Software Engineering Intern* **Jan - Aug 2026**

- Owned development of an AI-enabled clinical-trial intelligence platform integrating Microsoft Copilot Studio, RAG, Azure Functions, an MCP server, and ClinicalTrials.gov for large-scale study retrieval, synthesis, and comparison.
- Built cloud workflows and reusable API tools to query, filter, standardize, and version clinical-trial records, translating stakeholder questions into repeatable multi-study analyses for research, clinical, and commercial teams.
- Automated clinical-study design generation through Power Automate, replacing an outsourced vendor process with a validated internal pipeline and eliminating approximately $800K in external labor costs.
- Designed role-aware prompt classification, tool routing, and persistent memory by audience and analytical intent, enabling relevant multi-turn outputs across complex clinical-development workflows.

Aptar Pharma, Boonton, NJ, *Particle Engineering Intern* **May 2025 - Aug 2025**

- Developed a multivariate feature engineering dataset spanning feed viscosity, atomization energy, droplet size, particle-size distribution, powder yield, formulation composition, and stability to characterize formulation–process relationships.
- Applied model predictions to define operating conditions for scale-up to 300 g/hr, achieving a 16× throughput increase while maintaining target particle-size distributions for peptides, proteins, and encapsulated RNA.
- Used computational modeling alongside experimental design to reduce iterative formulation testing and identify excipient/process combinations suitable for pulmonary and nose-to-brain drug delivery.

Cornell DEBUT, Ithaca, NY, *Product Manager* **Oct 2023 - Present**

- Led an interdisciplinary engineering team through product definition, device development, computational integration, testing, and validation for MeyeAttention, a wearable pupillometry platform for neurological screening.
- Trained convolutional neural network (CNN) and logistic regression models for MeyeAttention, a machine-learning-based pupillary diagnostic designed to identify neurological signatures associated with ADHD, TBI, and other CNS conditions.
- Developed the computational pipeline connecting pupillometry data acquisition, feature processing, and ML-based classification, translating physiological signals into quantitative diagnostic outputs.
- Engineered pupillometry-derived biomarkers capturing norepinephrine-linked P300 responses under tonic and phasic visual stimuli to characterize neurological response patterns.
- Won 1st Place Overall at the 2025 BMES Medtronic Student Design Competition for the MeyeAttention diagnostic platform.

PROJECTS

PreSim, Ithaca, NY, *Independent Project* **Sep 2026 - Present**

- Developing a React-based chemical engineering learning platform that guides students from process concepts through flowsheet construction, material balances, and preparation for Aspen/ChemCAD simulation.
- Designing an LLM-powered backend to translate natural-language descriptions into structured process specifications and provide guidance on missing inputs, assumptions, and modeling decisions.
- Defining engineering constraints and benchmark cases based on conservation laws, unit consistency, and explicit modeling assumptions to evaluate process specifications and material-balance calculations.

Cornell Heat and Power Plant, Ithaca, NY, *Student Consulting Project* **Sep - Dec 2025**

- Evaluated combined heat and power plant operations through a chemical engineering consulting project, applying thermodynamics, fluid mechanics, and heat-transfer principles to assess system performance.
- Analyzed plant operating conditions and ambiguous datasets to identify opportunities for improved energy efficiency, resource utilization, and overall process optimization.

SPECIALIZED SKILLS

Technical: Python, SQL, Java, Node.js, Git/GitHub, CRUD APIs, Azure Functions, AWS, Microsoft Power Apps, Power Automate, Claude Code, Product Management, Machine Learning, Data Pipelines, Process Automation
Interests: Ultra-running, Pickup Basketball, AI, Pasta-making, Woks, Longevity, Creative Writing, Football